

A constructive proof that a Bézout domain need not be an elementary divisor domain

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Abstract

We give a constructive proof of the theorem of Hägg and Mörtberg, with GPT-6 Astra: there is a Bézout domain containing $\mathbb{Q}[x, y]$ over which the matrix $\begin{pmatrix} 1+x & y \\ y & 1-x \end{pmatrix}$ has no Smith normal form. The proof is written in the style of [5]. The only principles used beyond intuitionistic logic are unique choice and Noetherianity in the sense of Richman and Seidenberg. Krull dimension, maximal ideals and compactness are not used.

This note is a constructive version, written by Claude, of the proof in [4] (<https://arxiv.org/abs/2609.35229>). It has not been reviewed by Hägg and Mörtberg. The accompanying Lean formalization, a fork of theirs, is at <https://github.com/coquand/bezout-constructive>.

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1 Statement and plan

Theorem 1.1 (Hägg–Mörtberg with GPT-6 Astra [4]). *There is a Bézout domain $R \supseteq \mathbb{Q}[x, y]$ over which the matrix*

$$M = \begin{pmatrix} 1+x & y \\ y & 1-x \end{pmatrix}$$

is not equivalent to a diagonal matrix. In particular M has no Smith normal form.

The ring is constructed as an increasing union $R = \bigcup_n A_n$ of finitely presented $\mathbb{Q}$ -algebras with $A_0 = \mathbb{Q}[x, y]$. Every pair of elements of R is enumerated. At stage n one pair (a, b) of A_n is written $(a, b) = c(a', b')$ with $\gcd(a', b') = 1$, and A_{n+1} is a ring in which (a', b') becomes principal. This step is a *principalisation*, done by repeating two moves (divisorial and torsor) until an invariant can no longer decrease.

The obstruction is real and topological. Put $\Delta = x^2 + y^2 - 1$. Then

$$\det M = -\Delta, \quad M^2 - 2M = \Delta I,$$

and on the circle $\Delta = 0$, written $(x, y) = (a^2 - b^2, 2ab)$ with $a^2 + b^2 = 1$, one has $M = 2^t(a, b)(a, b)$. So $\frac{1}{2}M$ projects onto the line of the half-angle, a Möbius band. The final argument (Section 11) shows that a diagonal form over some A_N would give a nowhere-vanishing section of this band along a closed path of real points.

What has to be made constructive. The plan above is that of [4]. Every step uses classical notions: maximal ideals, the maximum of an invariant over them, minimal primes, Krull dimension, compactness of real points, and factoriality. We replace them as follows.

- Factoriality becomes *discrete gcd domain* (Section 4).
- Smoothness and dimension are *certificates*: explicit Jacobian covers (Section 5).
- Maximal ideals become *points* with values in number fields (Section 6).
- Noetherianity in the sense of Richman–Seidenberg (Section 3) gives termination of searches without a priori bounds.
- Compactness becomes a *chain lifting property* for finite chains of real points (Section 10).

2 Conventions

We work in Bishop-style constructive mathematics, as in [5].

- A ring is *discrete* if equality is decidable. An ideal is *detachable* if membership is decidable. A ring is *strongly discrete* if every finitely generated ideal is detachable. Finitely presented $\mathbb{Q}$ -algebras are strongly discrete and coherent (Gröbner bases, [5, Chapter VII]).
- A *discrete field* is a discrete ring in which every element is 0 or invertible. A *number field* is a finite-dimensional $\mathbb{Q}$ -algebra that is a discrete field.

- Real numbers are Cauchy sequences with modulus. We use: if $a < b$ then $x < b$ or $a < x$ (cotransitivity); inverses of positive reals; n -th roots of positive reals; and the fixed point of a contraction with rate $\frac{1}{2}$. We never test a real number for zero.

Unique choice. If for every x there is a unique y with $P(x, y)$, then there is a function f with $P(x, f(x))$. In Bishop's mathematics this is part of the notion of function. The construction of the tower is a recursion in which each stage is the unique object satisfying a decidable specification (canonical choices, Section 8). So it is a function by unique choice, and no countable or dependent choice is used.

3 Noetherianity and Krull's intersection theorem

We use Noetherianity in the sense of Richman and Seidenberg [5, II-3.2]: every ascending sequence of finitely generated ideals has two equal consecutive terms. Equivalently, for every sequence (a_k) there is j with $a_j \in \langle a_0, \dots, a_{j-1} \rangle$.

Proposition 3.1. *Finitely generated $\mathbb{Q}$ -algebras, their quotients and localisations, and the Rees algebras of their finitely generated ideals (Definition 3.3) are Noetherian.*

Proof. Hilbert's basis theorem in the constructive form of [3]. To pass to quotients and localisations without countable choice, one uses an inductive formulation of Noetherianity; see Appendix A. $\square$

The proofs below all follow one pattern. A process produces a sequence step by step. Noetherianity gives an index where the new term lies in the ideal of the earlier ones, and this contradicts the hypothesis that keeps the process running. So the process stops, although no bound is known in advance.

Lemma 3.2 (order along a non-unit). *Let D be a Noetherian domain with decidable divisibility by s , where $s \in D$ is neither 0 nor a unit. Every $a \neq 0$ can be written $a = s^k a'$ with $s \nmid a'$, and k is computed.*

Proof. Put $a_0 = a$. Put $a_{i+1} = a_i/s$ if $s \mid a_i$ (decidable), and $a_{i+1} = a_i$ otherwise. By Noetherianity there is $j \geq 1$ with $a_j \in \langle a_0, \dots, a_{j-1} \rangle$. If s divided $a_0, \dots, a_{j-1}$, this ideal would be $\langle a_{j-1} \rangle$ and $a_{j-1} = sa_j$. Then $a_j = cs a_j$, so $(1 - cs)a_j = 0$. Since s is not a unit, $1 - cs \neq 0$, so $a_j = 0$ and hence $a = 0$, a contradiction. So the division stops at some $i < j$, which is found by a finite check. $\square$

Definition 3.3 ([5, XII-2, before Fact 2.5]). Let $\mathfrak{a}$ be an ideal of a ring B and t an indeterminate. The *Rees algebra* of $\mathfrak{a}$ is the subalgebra

$$B[\mathfrak{a}t] = B \oplus \mathfrak{a}t \oplus \mathfrak{a}^2 t^2 \oplus \dots \subseteq B[t].$$

So a polynomial $\sum g_j t^j$ lies in $B[\mathfrak{a}t]$ iff $g_j \in \mathfrak{a}^j$ for every j . If $\mathfrak{a} = \langle a_1, \dots, a_k \rangle$, then $B[\mathfrak{a}t]$ is generated by $a_1 t, \dots, a_k t$, so it is a quotient of $B[T_1, \dots, T_k]$.

Lemma 3.4 (Krull step, after Perdry [6]). *Let $\mathfrak{a}$ be an ideal of a ring B . Suppose that in the Rees ring $B[\mathfrak{a}t]$ a relation*

$$b t^{m+1} = \sum_{i \leq m} g_i b t^i, \quad g_i \in B[\mathfrak{a}t],$$

holds. Then $b \in b \mathfrak{a}$. If moreover $1 + \mathfrak{a} \subseteq B^\times$, then $b = 0$.

Proof. Compare the coefficients of t^{m+1} . The coefficient of t^{m+1-i} in g_i lies in $\mathfrak{a}^{m+1-i} \subseteq \mathfrak{a}$, so $b = cb$ with $c \in \mathfrak{a}$, and then $(1 - c)b = 0$. $\square$

Corollary 3.5 (Krull's intersection theorem). *Let B be a local ring with maximal ideal $\mathfrak{m}$, whose Rees algebra $B[\mathfrak{m}t]$ is Noetherian, and in which membership in each $\mathfrak{m}^k$ is decidable. For $b \in B$ one computes either k with $b \notin \mathfrak{m}^k$, or a proof that $b = 0$.*

Proof. Test $b \in \mathfrak{m}^k$ for $k = 0, 1, \dots$. As long as the tests succeed, $bt^k \in B[\mathfrak{m}t]$. Noetherianity applied to the sequence (bt^k) gives the relation of Lemma 3.4, and then $b = 0$, because $1 + \mathfrak{m} \subseteq B^\times$. $\square$

4 Discrete gcd domains

Definition 4.1. A *discrete gcd domain* is a discrete domain with decidable divisibility in which every two elements have a gcd.

This replaces factoriality. It is all we need to write any pair as $(a, b) = c(a', b')$ with a', b' without common non-unit factor.

Lemma 4.2. $\mathbb{Q}[x, y]$ is a discrete gcd domain. If D is a discrete gcd domain, so are $D[X]$ and every localisation of D at a multiplicative set with decidable membership.

Proof. Gauss's lemma, and gcds over the discrete field of fractions [5, III]. For $\mathbb{Q}[x, y]$ one can also use Kronecker factorisation. $\square$

Lemma 4.3 (Nagata at a prime element). *Let D be a strongly discrete Noetherian domain and $s \in D$ a prime element, that is, D/sD is a domain. If $D[1/s]$ is a discrete gcd domain, so is D .*

Proof. Let $a, b \neq 0$. By Lemma 3.2 write $a = s^k a'$ and $b = s^l b'$ with $s \nmid a', b'$. Let $g \in D$ represent a gcd of a', b' in $D[1/s]$; dividing out powers of s (Lemma 3.2 again), we may assume $s \nmid g$.

- $g \mid a'$ in D . Indeed $s^N a' = gx$ for some N and $x \in D$. Since s is prime and $s \nmid g$, $s^N \mid x$, so $a' = g(x/s^N)$.
- A common divisor $c \in D$ of a', b' divides g in D . It divides g in $D[1/s]$, and $s \nmid c$ (because $s \nmid a'$), so the same argument applies.

Hence $s^{\min(k, l)} g$ is a gcd of a, b in D . The factor $s^{\min(k, l)}$ is correct because s is prime and $s \nmid a', s \nmid b'$. $\square$

Lemma 4.4 (generic linear form). *Let B be a discrete gcd domain and $c_0, \dots, c_\ell \in B$ with $\gcd(c) = 1$. Over $B' = B[\lambda_0, \dots, \lambda_\ell]$ the element $\pi' = \sum \lambda_i c_i$ is prime.*

Proof. B' is a discrete gcd domain. π' has degree 1 in the λ 's and content $\gcd(c) = 1$. In a factorisation one factor has degree 0 and divides every coefficient, so it is a unit, and π' is irreducible. In a discrete gcd domain an irreducible element π is prime. If $\pi \mid ab$ and $\pi \nmid a$, then $\gcd(\pi, a) = 1$, hence $\pi \mid b$. $\square$

For a ring B and $c = (c_0, \dots, c_\ell)$ put

$$J_B(c) = B[\sigma_0, \dots, \sigma_\ell] / \left(\sum c_i \sigma_i - 1 \right),$$

the *forcing algebra* of c .

Lemma 4.5. *If D is a domain and some $c_i \neq 0$, then $J_D(c)$ is a domain.*

Proof. The polynomial $f = \sum c_i \sigma_i - 1$ has content (1). By the Dedekind–Mertens lemma, $J_D(c)$ is torsion-free over D , so it embeds in $K[\sigma]/(f)$ with $K = \text{Frac } D$. There c_i is invertible, σ_i can be solved, and what remains is a polynomial ring over K . $\square$

Proposition 4.6. *Let B be a finitely presented $\mathbb{Q}$ -algebra that is a discrete gcd domain, and let $c_0, \dots, c_\ell \in B$ have no common non-unit divisor, with $c_0 \neq 0$. Then $J_B(c)$ is a discrete gcd domain.*

Proof. Let $B' = B[\lambda_0, \dots, \lambda_\ell]$ and $\pi' = \sum \lambda_i c_i$, which is prime by Lemma 4.4. The substitution $\tau_i = \sigma_i + \lambda_i \sigma'$ gives

$$J_B(c)[\lambda, \sigma'] \cong E = B'[\tau, \sigma'] / \left(\sum c_i \tau_i - \pi' \sigma' - 1 \right).$$

- π' is prime in E : $E/\pi' E \cong J_{B'/\pi'}(\bar{c})[\sigma']$, which is a domain by Lemma 4.5, because B'/π' is a domain and $\bar{c}_0 \neq 0$ ($\pi' \nmid c_0$ for degree reasons).
- Inverting π' solves σ' , so $E[1/\pi']$ is a polynomial ring over $B'[1/\pi']$, hence a discrete gcd domain (Lemma 4.2).
- E is finitely presented, hence Noetherian with decidable divisibility (Gröbner bases).

So E is a discrete gcd domain by Lemma 4.3. Gcds of elements of $J_B(c)$ computed in the polynomial ring $E \cong J_B(c)[\lambda, \sigma']$ have degree 0, so they are gcds in $J_B(c)$. $\square$

In the torsor step (Section 9), Lemma 4.3 is applied once more, at the element s .

5 Smoothness certificates

Let $A = \mathbb{Q}[Y_1, \dots, Y_N]/(G)$ be finitely presented.

Definition 5.1. A *certificate of dimension n* for A consists of the following data.

- Polynomials $h_1, \dots, h_r$ and $u_1, \dots, u_r$, and an integer $C > 0$, with $C - \sum u_k h_k \in (G)$. So the h_k are comaximal in A .
- For each k , polynomials $F = (F_1, \dots, F_N)$ with $F_1, \dots, F_{N-n} \in (G)$, together with U and exponents e, f , such that modulo (G)

$$\det\left(\frac{\partial F}{\partial Y}\right)U \equiv h_k^e \quad \text{and} \quad h_k^f G \subseteq (F_1, \dots, F_{N-n}).$$

The last n components $F_{N-n+1}, \dots, F_N$ are the *coordinates* of the piece.

So on $D(h_k)$ the ideal (G) is generated by $N - n$ equations, and their Jacobian, completed by the coordinates, is invertible.

Lemma 5.2 (standard form of a piece). *Write $h = h_k$. Then*

$$A[1/h] \cong \mathbb{Q}[Y, Z_1, \dots, Z_n, t] / (F_1, \dots, F_{N-n}, F_{N-n+j} - Z_j, ht - 1),$$

and the Jacobian of this presentation with respect to (Y, t) is a unit. Hence $A[1/h]$ is étale over $\mathbb{Q}[Z_1, \dots, Z_n]$, and $\Omega_{A[1/h]/\mathbb{Q}}$ is free with basis $dZ_1, \dots, dZ_n$.

Proof. The isomorphism is given by two explicit inverse maps. No kernel has to be computed. The Jacobian is $\det(\partial F/\partial Y) \cdot h$, which is a unit modulo the relations. $\square$

From Lemma 5.2 we get explicit local coordinates, explicit dual derivations $\partial/\partial Z_j$, and Newton lifting of solutions at points. These are all the smoothness properties we use; no general smoothness criterion is needed.

Remark (smooth versus smooth of constant dimension). A certificate says more than “ A is smooth over $\mathbb{Q}$ ”: it fixes the rank n of $\Omega_{A/\mathbb{Q}}$ on every piece. For one equation F , the condition that $\{F = 0\}$ is a smooth hypersurface,

$$1 \in \left(F, \frac{\partial F}{\partial Y_1}, \dots, \frac{\partial F}{\partial Y_N}\right),$$

is a certificate of dimension $N - 1$, with $h_k = \partial F/\partial Y_k$. Smoothness alone allows the rank to vary (for instance $X^2 = I$ in M_n , which has isolated points). In our tower all rings are discrete domains, whose only idempotents are 0 and 1, so the rank is automatically constant. The certificate records it.

Proposition 5.3 (propagation). *1. If B has a certificate of dimension n and $c_0, \dots, c_r \in B$, then $J_B(c)$ has a certificate of dimension $n + r$.*

2. If A has a certificate of dimension n , the extended Rees algebra of a weighted centre (defined in Section 9) has a certificate of dimension $n + 1$.

Proof. 1. The c_i are comaximal in $J = J_B(c)$, since $\sum c_i \sigma_i = 1$ there. On $D(c_i)$ one solves $\sigma_i: J[1/c_i] \cong B[1/c_i][\sigma_j : j \neq i]$. The pieces are the products $c_i h_k$, and the coordinates are those of B together with the σ_j , $j \neq i$. In the language of the remark, $\sum c_i \sigma_i - 1$ is a smooth hypersurface relative to B , because the derivatives $\partial/\partial \sigma_i = c_i$ generate (1) in J . 2. On a chart where the centre is $(x_1, \dots, x_k)$ with integer weights w_i ,

$$\mathcal{R} \cong A[s, u_1, \dots, u_k]/(x_i - s^{w_i} u_i),$$

and the equations $x_i - s^{w_i} u_i$ are solved for x_i , which are coordinates of A . The new coordinates are those of A with x_i replaced by u_i , plus s . $\square$

The certificate gives a dimension without Krull dimension. All rings of the tower have dimension $n \geq 2$. The dimension n bounds the length of the invariants: $k \leq n$ (Section 7).

6 Points

Definition 6.1. A *point* of a finitely presented $\mathbb{Q}$ -algebra A is a $\mathbb{Q}$ -algebra map $q : A \rightarrow L$, where L is a number field (Section 2). Its kernel $\ker q$ is a detachable prime, and its residue field $\kappa(q) \subseteq L$ is a discrete field.

A point of $\mathbb{Q}[Y]/(G)$ is a solution of $G = 0$ in L^N . In the classical proof, every statement “for every maximal ideal $\mathfrak{m} \supseteq I$ ” becomes “for every point q with $I \subseteq \ker q$ ”. Classically the two are equivalent by the Nullstellensatz. Constructively we use the following.

Proposition 6.2 (Nullstellensatz). *Let $\mathfrak{p}$ be a finitely generated prime of A and $a \notin \mathfrak{p}$. There is a point q with $\mathfrak{p} \subseteq \ker q$ and $q(a) \neq 0$. More generally, if I is a finitely generated ideal with $1 \notin I$, there is a point with $I \subseteq \ker q$.*

Proof. A/I is a nonzero finitely presented $\mathbb{Q}$ -algebra (decided by Gröbner bases). Noether normalisation and the primitive element theorem produce a quotient that is a finite field extension [5, VII]. For the first claim, apply this to $(A/\mathfrak{p})[1/a]$. $\square$

Lemma 6.3 (transfer). *1. Membership in $\ker q$ is decidable. Hence a negative statement of the form $x \notin \ker q$ may be proved by contradiction.*

2. If $\varphi : A \rightarrow U$ is a $\mathbb{Q}$ -algebra map and Q a point of U , then $Q \circ \varphi$ is a point of A . This replaces the classical fact that the contraction of a maximal ideal of a finitely generated algebra is maximal.

3. (Local-global) If $g_1, \dots, g_m$ are comaximal and a property of finitely presented modules or ideals holds on each $D(g_i)$, it holds on A [5, II].

Proposition 6.4 (covering the zero set). *Let I be a finitely generated ideal of A . Suppose that for every point q with $I \subseteq \ker q$ we can compute $g_q \notin \ker q$. Then finitely many points $q_1, \dots, q_m$ satisfy $1 \in I + (g_{q_1}, \dots, g_{q_m})$.*

Proof. Build the list $g_{q_1}, g_{q_2}, \dots$. If $1 \in I + (g_{q_1}, \dots, g_{q_j})$ (decidable) stop. Otherwise Proposition 6.2 gives a point q_{j+1} of $V(I + (g_{q_1}, \dots, g_{q_j}))$. Then $g_{q_{j+1}} \notin I + (g_{q_1}, \dots, g_{q_j})$, because q_{j+1} kills the latter. Continue the sequence by 1 after a stop, and apply the Noetherianity of A/I (Proposition 3.1). It gives j with $g_{q_{j+1}} \in I + (g_{q_1}, \dots, g_{q_j})$. By the last observation, the process has stopped by step j . $\square$

This is our substitute for quasi-compactness of $\text{Spec } A$.

7 The invariant at a point

7.1 Centred charts and Taylor expansion

Let q be a point of A , lying in a piece $D(h)$ of the certificate, and let $Z_1, \dots, Z_n$ be the coordinates of the piece. After a translation by the values $q(Z_j)$ (and, when $\kappa(q) \neq \mathbb{Q}$, after adjoining the residue field), the coordinates vanish at q ; such a chart is *centred* at q . Let A_q be the localisation at $\ker q$, and $\mathfrak{m}_q$ its maximal ideal. Lemma 5.2 gives the *Taylor map*

$$\tau : A_q \rightarrow \kappa(q)[[Z_1, \dots, Z_n]],$$

obtained by Newton lifting. Its coefficients are computed. By Corollary 3.5 it is injective. Moreover $\mathfrak{m}_q^k$ consists exactly of the elements whose expansion has no terms of degree $< k$.

Lemma 7.1 (order at a point). *For $f \in A_q$ we can decide whether $f = 0$, and if $f \neq 0$ compute $\text{ord}_q(f)$, the largest k with $f \in \mathfrak{m}_q^k$. The same holds for a finitely generated ideal I , as the minimum over its generators.*

Proof. Test the homogeneous parts of $\tau(f)$ of degree $0, 1, 2, \dots$; each test is a finite computation over $\kappa(q)$. Corollary 3.5 says that this search ends, either at a nonzero part or with a proof that $f = 0$. No bound is known in advance, and no Markov principle is used. $\square$

Weighted orders work in the same way. For rational weights $e_1, \dots, e_k \geq 0$ on some coordinates, f lies in the filtration level $\mathcal{F}_a$ if and only if $\tau(f)$ has no monomial Z^β with $\sum e_i \beta_i < a$. In particular, the associated graded ring $\bigoplus \mathcal{F}_a / \mathcal{F}_{>a}$ embeds in weighted-homogeneous series over $\kappa(q)$, so it is a domain.

7.2 The invariant

At a point $q \supseteq I$ we use the invariant of Abramovich–Temkin–Włodarczyk [1], in the differential presentation of Brais [2] (“Method 1”). It is a finite sequence

$$\text{inv}_q(I) = (a_1, \dots, a_k), \quad k \leq n,$$

of positive rationals, computed as follows.

- $a_1 = \text{ord}_q(I)$ (Lemma 7.1).
- A *maximal contact* element is a derivative of order $a_1 - 1$ of a generator of order a_1 . It has order 1 at q , and it is found by a finite search.
- The next entries come from the coefficient ideal on the maximal contact hypersurface. Each is a minimum over a finite, explicitly bounded box of multi-indices. It has the form $b/D!$ with $b \in \mathbb{N}_{>0}$, where D is the product of the denominators of the previous entries.

Equivalently one may record the weights $w_i = 1/a_i$. Then maxinv corresponds to the lexicographically *least* weight vector. Every step is a finite computation over the discrete field $\kappa(q)$. The same data give a *marked centre* at q : coordinates $x_1, \dots, x_k$ and weights $w_1, \dots, w_k$. Its admissibility ($I \subseteq \mathcal{F}_d$ for the appropriate d) is decided coefficientwise.

Lemma 7.2 (well-ordering). *Order invariants lexicographically, with the convention of [1] for sequences of different lengths. Restricted to sequences of length $\leq n$ that arise as invariants, this order is well founded: well-founded induction and recursion are valid on it (Appendix A).*

Proof. Induction on the length. Once $a_1, \dots, a_{i-1}$ are fixed, the next entry is $b/D!$ with D fixed by the previous entries. So it ranges over $\frac{1}{D!}\mathbb{N}$, which is well founded by induction on $\mathbb{N}$. The lexicographic product of well-founded orders is well founded, by nested induction. $\square$

Proposition 7.3 (upper semicontinuity). *For every point $q \supseteq I$ there is $g \notin \ker q$ such that $\text{inv}_{q'}(I) \preceq \text{inv}_q(I)$ for every point $q' \supseteq I$ of $D(g)$; the computation runs at every point of $D(g)$. Moreover, on $D(g)$ the points with $\text{inv}_{q'}(I) = \text{inv}_q(I)$ are those of $V(x_1, \dots, x_k)$, for the marked centre at q .*

Sketch. All the data of the computation at q (a nonzero coefficient, a unit derivative, an admissible marked centre) are elements that do not vanish at q . Let g be their product. At a point q' of $D(g)$ the same data are still available, so the computation at q' cannot produce a larger invariant. It produces the same invariant exactly when the marked centre vanishes at q' . Each inequality is a statement about non-membership in $\ker q'$, which is decidable (Lemma 6.3). $\square$

8 The maximal locus and its components

Proposition 8.1. *For a finitely generated ideal I with $0 \neq I \neq (1)$, one can compute $\text{maxinv}(I)$, a point q_0 where it is attained, and the property*

$$\text{inv}_q(I) \preceq \text{maxinv}(I) \quad \text{for every point } q \supseteq I.$$

Proof. Combine Proposition 7.3 with Proposition 6.4. This gives finitely many points $q_1, \dots, q_m$ whose neighbourhoods $D(g_{q_j})$ cover $V(I)$. Take the maximum of the $\text{inv}_{q_j}(I)$. Every point $q \supseteq I$ lies in some $D(g_{q_j})$, decidable, so $\text{inv}_q(I) \preceq \text{inv}_{q_j}(I)$. $\square$

The *locus ideal* L is obtained by gluing: on the $D(g_{q_j})$ with $\text{inv}_{q_j} = \text{maxinv}$ it is the marked centre ideal, and elsewhere it is (1) . Its points are exactly the points where the maximum is attained. Since the centre is given by part of a coordinate system, $V(L)$ is smooth.

Components. The *components* are the minimal primes of $\sqrt{L}$. They are computed with Perdry's strongly Noetherian theory [7]: a primality test by induction on the number of variables, and a tree of prime-or-split decisions whose finiteness again comes from Noetherianity. Different components $\mathfrak{p} \neq \mathfrak{p}'$ are comaximal. Indeed, $1 \in \mathfrak{p} + \mathfrak{p}'$ is decidable. If it fails, Proposition 6.2 gives a point q of $V(\mathfrak{p} + \mathfrak{p}')$. Near q the locus is $V(x_1, \dots, x_k)$ for a centred chart, and these coordinates generate a prime. Both $\mathfrak{p}$ and $\mathfrak{p}'$ are minimal over the locus, so both are equal to this prime, and $\mathfrak{p} = \mathfrak{p}'$.

Canonical choice. Each finitely presented ring of the tower comes with an explicit enumeration. Order the finite lists of generators by first occurrence in that enumeration. The component treated next is the one whose canonical list of generators is least. Together with the canonical point q_0 (least in the same sense), all choices are canonical, so the step is a function by unique choice.

Remark. One could try a dynamical treatment: keep a radical centre and split it only when needed. This does not work for the tower. With $r \geq 2$ components, the forcing algebra has class group $\mathbb{Z}^r / \mathbb{Z}(1, \dots, 1) \neq 0$. A split discovered later would therefore invalidate a stage that has already been used.

9 The principalisation step

Let $I = (a', b')$ in a stage A with a certificate of dimension n , let $\text{maxinv}(I)$ have first entry a_1 , and let $\mathfrak{p}$ be the chosen component, with weighted centre $x_1^{1/w_1}, \dots, x_k^{1/w_k}$ along $\mathfrak{p}$.

9.1 Divisorial step ($k = 1$)

Then $\mathfrak{p}$ is principal. Let π be a gcd of a generating family of $\mathfrak{p}$. At one point of $V(\mathfrak{p})$, $\mathfrak{p}$ is locally principal, and gcds of families are preserved by localisation, so $\mathfrak{p} = (\pi)$ and π is prime. The weight is $1/a$, and $I = \pi^a I_1$. The new ideal is I_1 , in the same ring.

Lemma 9.1. *No point of $V(\pi)$ contains I_1 . So the component $\mathfrak{p}$ disappears from the locus of I_1 , and the invariant does not increase elsewhere.*

Proof. Let $q \supseteq I_1$ with $\pi \in \ker q$. Then $I = \pi^a I_1 \subseteq \mathfrak{m}_q^{a+1}$, so the first weight at q would be at most $1/(a+1)$, contradicting the maximality of $\text{maxinv}(I)$, whose first entry is a . Membership in $\ker q$ is decidable, so this negative argument is valid (Lemma 6.3). $\square$

9.2 Torsor step ($k \geq 2$)

We need two variants of Definition 3.3: the filtration is weighted, and negative degrees are allowed. Let $\mathcal{F}_j$ ($j \in \mathbb{Z}$) be the *weighted filtration* of the centre. On a chart where the centre has coordinates $x_1, \dots, x_k$ and integer weights $w_1, \dots, w_k$, $\mathcal{F}_j$ is the ideal generated by the monomials x^β with $\sum w_i \beta_i \geq j$; in particular $\mathcal{F}_j = A$ for $j \leq 0$, and $\mathcal{F}_i \mathcal{F}_j \subseteq \mathcal{F}_{i+j}$. These local ideals glue. The *extended Rees algebra* is the subalgebra

$$\mathcal{R} = \bigoplus_{j \in \mathbb{Z}} \mathcal{F}_j T^j \subseteq A[T, T^{-1}], \quad s = T^{-1} \in \mathcal{R}.$$

On the chart, $\mathcal{R} = A[s, x_1 T^{w_1}, \dots, x_k T^{w_k}]$. Putting $u_i = x_i T^{w_i}$ gives $x_i = s^{w_i} u_i$, which is the presentation used in Proposition 5.3. Moreover $\mathcal{R}/s\mathcal{R} \cong \bigoplus_{j \geq 0} \mathcal{F}_j / \mathcal{F}_{j+1} = \text{gr } \mathcal{F}$, the associated

graded ring, and $\mathcal{R}[1/s] = A[T, T^{-1}]$. Choose $\pi \in \mathcal{F}_1 \setminus \mathcal{F}_2$, and let $y = (y_0, \dots, y_\ell) = (\pi T, h_1, \dots, h_\ell)$, where the h_i generate $\mathcal{R}_+$ (in practice $x_i T^{w_i}$). Put

$$U = J_{\mathcal{R}}(y) = \mathcal{R}[\sigma_0, \dots, \sigma_\ell] / \left(\sum y_i \sigma_i - 1 \right).$$

The weights of the centre are $v_i = w_i/d$ for a common denominator d , and admissibility gives $I \subseteq \mathcal{F}_d$. Every $f \in I$ satisfies $f = s^d(fT^d)$, so

$$IU = s^d I_1 U, \quad I_1 = (fT^d : f \in I) \subseteq \mathcal{R},$$

where I_1 is the *weak transform* of I . By Proposition 5.3, U has a certificate of dimension $n + 1 + \ell$.

Proposition 9.2. *s is a prime element of U , and U is a discrete gcd domain.*

Proof. U/sU is the forcing algebra of the images of the y_i over $\mathcal{R}/s\mathcal{R} = \text{gr } \mathcal{F}$, and the image of πT is nonzero. By Lemma 4.5 it is enough that $\text{gr } \mathcal{F}$ is a domain. That is checked at *one* point q_0 of $V(\mathfrak{p})$: by Section 7, $\text{gr } \mathcal{F}$ embeds into weighted-homogeneous series over $\kappa(q_0)$. A statement $z \notin \mathcal{F}_{j+1}$ is transferred from q_0 to the whole component by decidability. Then $U[1/s] = J_B(y)$ with $B = A[T, T^{-1}]$. Here $y_0 = \pi T \neq 0$, and the y_i have no common non-unit divisor in B : up to units they are generators of $\mathcal{F}_1 \supseteq \mathfrak{p}$, and $k \geq 2$. So $U[1/s]$ is a discrete gcd domain (Proposition 4.6). Lemma 4.3 concludes. $\square$

Proposition 9.3 (the invariant drops). *For every point Q of U with $I_1 \subseteq \ker Q$, $\text{inv}_Q(I_1) \prec \text{maxinv}(I)$, or Q lies over a component $\mathfrak{p}' \neq \mathfrak{p}$, where the invariant is unchanged.*

Sketch. Let $q = Q|_A$ (Lemma 6.3). There are two cases.

- *Away from the centre.* If $q \notin V(\mathfrak{p})$, then U is locally a polynomial ring over A , and nothing changes. The other components lift to the primes $\mathfrak{p}'U$, which are comaximal with $\mathfrak{p}U$. So the locus ideal of I_1 is $\bigcap_{\mathfrak{p}' \neq \mathfrak{p}} \mathfrak{p}'U$, which has one component less.
- *Over the centre,* with $s \in \ker Q$; then $\mathcal{R}_+ \not\subseteq \ker Q$, since the y_i generate (1) in U . Let Q_0 be the *vertex point* of $\mathcal{R}$: the point $\sum f_j T^j \mapsto q_0(f_0)$, which lies in $V(\mathcal{R}_+)$. At Q_0 the weak transform I_1 has invariant $\text{maxinv}(I)$, attained along the Rees centre, which lies in $V(\mathcal{R}_+)$. On a neighbourhood $D(G)$ of Q_0 , Proposition 7.3 bounds the invariant by $\text{maxinv}(I)$, with equality only on the Rees centre. Hence $\text{inv}_Q(I_1) \prec \text{maxinv}(I)$ as soon as Q restricts to a point of $D(G)$. To arrange this we use the $\mathbb{Q}^\times$ -action $T \mapsto \mu T$, which preserves I_1 up to a unit. $Q(\sigma_\mu G)$ is a polynomial in μ with coefficients in the number field of Q , of some degree e , and its constant term $Q(G_0)$ is nonzero. So one of $\mu = 1, \dots, e + 1$ is not a root, and it is found by the zero test. $\square$

Theorem 9.4 (principalisation). *Let A be a stage with a certificate, (a', b') without common non-unit factor. A finite sequence of divisorial and torsor steps produces a stage $A' \supseteq A$ with a certificate, a discrete gcd domain, and $g \in A'$ with $(a', b')A' = gA'$. The construction is a function of (A, a', b') .*

Proof. The pair $(\text{maxinv}(I), \text{number of components})$ decreases lexicographically at each step: the divisorial step and the torsor step remove the chosen component, without increasing the invariant elsewhere. By Lemma 7.2 the order is well founded, so the steps are defined by well-founded recursion. The recursion stops when $I = (1)$. The accumulated factor π^a or s^d of each step generates the original ideal. Every choice is canonical (Section 8), so unique choice turns the recursion into data. $\square$

9.3 The tower

Let $A_0 = \mathbb{Q}[x, y]$, with the trivial certificate of dimension 2. Enumerate pairs so that every pair of elements of every A_n is treated at some later stage. At stage n , write the pair as $c(a', b')$ using the gcd of A_n , and apply Theorem 9.4. Then $R = \bigcup_n A_n$ is a domain. Every pair (a, b) becomes principal at some stage, and stays principal, so R is a Bézout domain. The maps $A_n \rightarrow A_{n+1}$ are injective, since the forcing algebra is faithful when $\text{Ann}(c) = 0$ [5, II.2.14] and the Rees algebra contains A .

10 Chain lifting

A *real point* of $A_n = \mathbb{Q}[Y]/(G)$ is a solution of $G = 0$ in $\mathbb{R}^N$; let $X_n \subseteq \mathbb{R}^N$ be the set of real points. A *closed ε -chain* is a finite sequence $P_0, \dots, P_m$ with $|P_{i+1} - P_i| < \varepsilon$ and $P_m = P_0$.

Definition 10.1. A map $\varphi : X' \rightarrow X$ has the *chain lifting property* if for every bound C there is a bound C' such that for every $\varepsilon > 0$ there is $\delta > 0$ with the following property. Every closed δ -chain $P_0, \dots, P_m$ in $X \cap [-C, C]^N$ has a closed ε -chain $P'_0, \dots, P'_{m'}$ in $X' \cap [-C', C']^{N'}$ and a map $\rho : \{0, \dots, m'\} \rightarrow \{0, \dots, m\}$ with

$$\rho(0) = 0, \quad \rho(m') = m, \quad \rho(j+1) - \rho(j) \in \{0, 1\}, \quad |\varphi(P'_j) - P_{\rho(j)}| < \varepsilon.$$

This is a finitary form of “ φ is a proper surjection with connected fibres”. It mentions only finitely many points at a time.

Lemma 10.2. *If φ and ψ have the chain lifting property and φ is Lipschitz on bounded sets, then $\varphi \circ \psi$ has the chain lifting property.*

Proof. Lift first along φ with a small ε_1 , then along ψ , and compose the reindexings. A Lipschitz constant K of φ on the relevant bounded set turns ε errors upstairs into $K\varepsilon$ errors downstairs. The parameters are chosen in the order C, C', C'' , and then ε, δ . $\square$

The projections of the tower are polynomial maps, so they are Lipschitz on bounded sets. It remains to treat one step. The divisorial step does not change the ring, and a principalisation is a finite composite of steps.

Proposition 10.3. *For a torsor step $A \rightarrow U$, the induced map $X_U \rightarrow X_A$ has the chain lifting property.*

Sketch. Let $\nu = \sum_i g_i^{E_i}$ be the weighted distance to the centre, with exponents making the terms weighted-homogeneous of a common degree D . Then two-sided bounds $cr^D \leq \nu \leq c'r^D$ hold in chart coordinates.

- *Away from the centre*, where $\nu > \eta$, the lift is unique and explicit: $s = \nu^{1/2D}$ (a positive root), and σ is determined by a fixed formula. Lifting point by point gives the chain.
- *Near the centre*, the fibre over a point is a weighted sphere $\{\varphi(u) = 1\}$ of dimension $k - 1 \geq 1$, which is connected. Consecutive lifted points are joined inside the fibre by relay paths made of three linear segments in chart coordinates. Charts are changed through ν .
- The analytic input is a polynomial inverse function theorem, whose radius is uniform in the base point on bounded sets. It is proved by a Kantorovich–Newton step, i.e. the fixed point of a contraction with rate $\frac{1}{2}$. It uses only inverses and roots of positive reals; there is no real division by a number not known to be positive, and no Markov principle. $\square$

Corollary 10.4. *For every N , the map $X_{A_N} \rightarrow X_{A_0} = \mathbb{R}^2$ has the chain lifting property.*

11 Proof of Theorem 1.1

Suppose $PMQ = \text{diag}(e_1, e_2)$ with P, Q invertible over R . Over a Bézout domain, $\text{diag}(e_1, e_2)$ is equivalent to $\text{diag}(g, e_1 e_2 / g)$ with $g = \gcd(e_1, e_2)$, so we may assume $e_1 \mid e_2$. All entries lie in some A_N . Then e_1 divides the entries of M , which generate (1) because $(1+x) + (1-x) = 2$. So e_1 is a unit. With (p, q) the first row of P , (u, v) the first column of Q and $\mu = e_1^{-1}$,

$$\mu(p, q) M \begin{pmatrix} u \\ v \end{pmatrix} = 1 \quad \text{in } A_N. \quad (11.1)$$

Half-angles. For $-1 \leq t \leq 1$ put

$$(a(t), b(t)) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right), \quad \gamma(t) = (a^2 - b^2, 2ab) \in \{\Delta = 0\}.$$

As t runs from -1 to 1 , γ runs once around the circle, from $(-1, 0)$ back to $(-1, 0)$, while (a, b) goes from $(0, -1)$ to $(0, 1)$. On the circle $M = 2^t(a, b)(a, b)$, so

$$M \begin{pmatrix} u \\ v \end{pmatrix} = 2l \begin{pmatrix} a \\ b \end{pmatrix}, \quad l = ua + vb.$$

A lower bound. On a bounded region the functions p, q, u, v, μ (polynomials in the coordinates of A_N) are bounded, and u, v are Lipschitz. By (11.1), at a real point of A_N lying over a point of the circle, $2\mu l(pa + qb) = 1$, so $|l| \geq 2c$ for some $c > 0$. At a real point P' lying ε -close to $\gamma(t)$, the same bound holds up to an error that tends to 0 with ε . So $|l(P', t)| \geq c$ once ε is small enough.

The chain. Choose $-1 = t_0 < \dots < t_m = 1$ so fine that $P_i = \gamma(t_i)$ is a closed δ -chain. Lift it by Corollary 10.4 to a closed ε -chain P'_j with reindexing ρ . Put $l_j = u(P'_j)a(t_{\rho(j)}) + v(P'_j)b(t_{\rho(j)})$. For small ε and δ :

- $|l_j| \geq c$ for all j ;
- $|l_{j+1} - l_j| < c$, because u, v are Lipschitz, $t \mapsto (a, b)(t)$ is 4-Lipschitz, and $\rho(j+1) - \rho(j) \in \{0, 1\}$.

By cotransitivity, applied to $-c/2 < c/2$, each l_j satisfies $l_j > c/2$ or $l_j < -c/2$, hence $l_j \geq c$ or $l_j \leq -c$. A sign change between consecutive values would give $|l_{j+1} - l_j| \geq 2c$, so two consecutive values have the same sign. So $l_{m'}$ and l_0 have the same sign. But $P'_{m'} = P'_0$, and $(a, b)(t_m) = -(a, b)(t_0)$, so $l_{m'} = -l_0$. This is a contradiction. Hence M has no diagonal form over R . $\square$

Comment. This last step is the constructive content of the intermediate value theorem used classically. It is where the ordered field $\mathbb{R}$ enters. Over a field in which -1 is a square, the circle ring is a principal ideal domain and there is no obstruction.

12 What is used

- Intuitionistic logic, unique choice, and inductive definitions (Appendix A).
- Constructive algebra from [5]: Gröbner bases, the Nullstellensatz, local-global principles, gcds over a discrete field, Kronecker factorisation, primitive elements.
- The Hilbert basis theorem in inductive form [3], and Perdry's theory of minimal primes [7].

- On $\mathbb{R}$: cotransitivity, inverses and roots of positive numbers, and the fixed point of a contraction with rate $\frac{1}{2}$.

No maximal ideals, Krull dimension, compactness, Markov principle or countable choice are used.

A Inductive definitions and bar induction

This appendix explains the inductive notions behind Proposition 3.1 and Lemma 7.2. They are not needed to follow the main text.

Inductive definitions. An inductive definition introduces a property by rules: the property is the smallest one closed under the rules. The natural numbers are generated by 0 and $n \mapsto n + 1$, and induction on $\mathbb{N}$ says exactly that every property closed under these rules holds for all n . The same is true for any inductive definition. This is *induction on the definition*. In type theory inductive definitions are primitive. In Bishop’s style they are accepted as generalised inductive definitions, as [5] does for the Krull dimension (Hdim).

Definition A.1 (accessibility). Let $\prec$ be a relation on a set E . The property $\text{Acc}(x)$ is defined inductively by one rule: if $\text{Acc}(y)$ for all $y \prec x$, then $\text{Acc}(x)$. The relation is *well founded* if $\text{Acc}(x)$ holds for every x .

Induction on Acc is well-founded induction, and it justifies well-founded recursion: a function may be defined at x from its values at all $y \prec x$. This is how the principalisation is defined (Theorem 9.4). The lexicographic product of two well-founded relations is well founded, by nested induction; this gives Lemma 7.2.

Bars. Let A be a commutative ring. Lists are written with the newest entry first.

Definition A.2 (inductive Noetherianity). The property $\text{Bar}(l)$ of a list l of elements of A is defined inductively by two rules.

1. If $a \in \langle l \rangle$, then $\text{Bar}(a :: l)$.
2. If $\text{Bar}(a :: l)$ for every $a \in A$, then $\text{Bar}(l)$.

A is *inductively Noetherian* if Bar holds for the empty list.

Read $\text{Bar}(l)$ as “however l is continued, it eventually gets an entry in the ideal of the older ones”. This form is due to Richman, Seidenberg, Jacobsson–Löfwall and Coquand–Persson [3].

Lemma A.3. *An inductively Noetherian ring is Noetherian in the sense of Richman–Seidenberg.*

Proof. By induction on $\text{Bar}(l)$, prove: “every sequence beginning with l has a term in the ideal of the earlier terms”. Under rule 1, the newest entry of l is such a term. Under rule 2, apply the induction hypothesis to l extended by the next term of the sequence. $\square$

Proposition A.4. 1. *A discrete field is inductively Noetherian.*

2. *If A is inductively Noetherian, so are $A/\mathfrak{a}$ and $S^{-1}A$.*

3. *$\mathbb{Q}[Y_1, \dots, Y_N]$ is inductively Noetherian [3].*

Sketch. 1. For a list l in a discrete field, decide whether $\langle l \rangle$ is 0 or (1). In the second case every extension is good. In the first case an extension a is 0 (good) or a unit, after which every extension is good. 2. Induction on Bar. At each use of rule 2, lift *one* element. 3. Dickson’s lemma is proved as an inductive bar on monomials, and Gröbner reduction transports it to ideals. $\square$

Comment. This is why the inductive form is used. A sequence in $A/\mathfrak{a}$ or $S^{-1}A$ cannot be lifted to a sequence in A without countable choice. In the inductive form only one element is lifted at a time. Proposition 3.1 follows from Proposition A.4, Lemma A.3, and the fact that a Rees algebra is a quotient of a polynomial ring.

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